N83 Koblitz: K1107 index calculus against precomputed-table rho

Run 4, `records/run-20261028T1511Z` · source `de31b7c99` · 2026-10-28 · L0 shared VM

1. Question and decision

Does the current N83 cold IC baseline (K1107/root11 dynamic, 2026-10-21) beat Pollard rho online once rho gets matched precomputation (a Bernstein–Lange table of G-only distinguished points)? Protocol: `PROTOCOL.md`, frozen before any run, with three amendments for runs 1–3, which failed on infrastructure and are kept, not combined.

Decision under the frozen rule (§4): precomputed rho is faster online in all three matched arms. Each 95% interval of the median paired ratio IC/precomputed-rho lies entirely above 1. This matches the frozen prediction. It is a reproduction diagnostic on this host only, not an improvement, gate result or leaderboard row.

2. Inputs and host

- Curve `EC1N83Ckb1a1haacb69ee9f37` ($y^2 + xy = x^3 + x^2 + 1$ over $\mathbb{F}_{2^{83}}$), r = 8569786107849059; the 16 public points of the 2026-10-16 control (sha256-checked).
- IC: `koblitz_orbit_dlp_fast construct:83:1:1107`, rank seed 29, root11 dynamic environment, `taskset -c 0-3`, one process for all targets.
- Rho arms: configs of the 2026-10-16 control; (a) 650 s table walks, 1 thread; (a′) walks budgeted at IC target-independent wall − 2 s = 149.6 s, 1 thread; (b) table bytes = IC max RSS (2,032,291,840 B), 4 threads; plain empty-table rho as positive control.
- Host: Intel Xeon @ 2.10 GHz, 4 vCPU, about 15 GB, no PMU. Load gate passed (1-min load 0.21 at start).
- Analysis: `n83_precomputed_rho_analyze` (exit 0): every recovered scalar replayed on the generic library curve; all arms agree; all match the hosted reference answers.

3. Online results (measured)

arm	threads	online median ms [95% CI]	median paired IC/arm [95% CI]	IC slower
IC K1107	4 / 1	115.0 [44.6, 379.7]		
precomputed (a) 650 s	1	3.69 [1.99, 4.87]	**40.9** [7.58, 141.9]	15/16
precomputed (a′) IC wall	1	15.2 [11.9, 30.7]	**4.10** [1.58, 24.8]	12/16
precomputed (b) IC RSS	4	4.21 [2.16, 9.39]	**33.1** [6.60, 74.4]	15/16
plain rho (control)	1	2,226 [1,561, 2,654]	rho/IC 10.7 [6.33, 88.8]	1/16

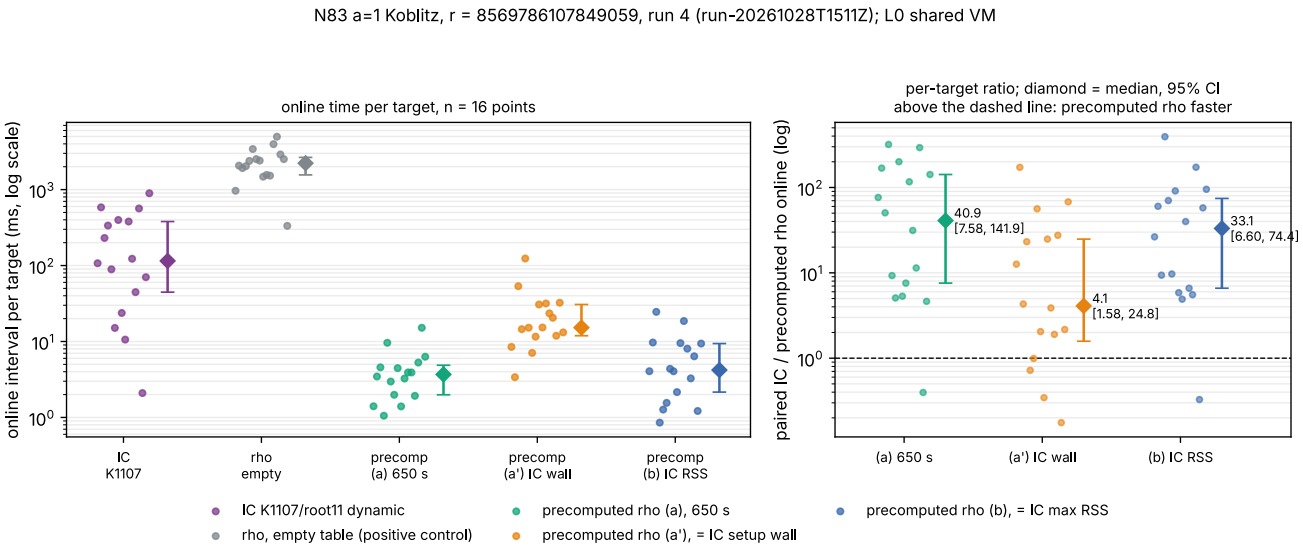


Figure 1: Run 4 (run-20261028T1511Z). Left: online interval per target for every arm. Right: per-target paired ratio IC/precomputed rho; diamonds are medians with 95% bootstrap intervals (10,000 target resamples). Values from `analysis.json`; layout by `plot.py`.

4. Cold cost (derived: setup + sum of 16 online intervals)

arm	setup s	online total s	cold s	peak RSS
IC (4 threads, 314 s user CPU)	151.6	3.87	155.5	2.03 GB
precomputed (a)	651.0	0.07	651.0	0.32 GB

precomputed (a′)	149.8	0.42	150.2	0.08 GB
precomputed (b) (4 threads)	586.2	0.11	586.3	2.04 GB
plain rho	0	36.96	36.96	

Reported beside the decision, not ranked. Plain rho solved all 16 targets in about a quarter of IC's whole-process wall on this host.

5. Operation counts (§5a)

Median paired IC/precomputed-rho count ratio at w = 0 / 0.1 / 1 GAE per unpriced native operation: (a) 0.0005 / 2.04 / 8.18; (a′) 0.0001 / 0.41 / 1.64; (b) 0.0009 / 3.76 / 15.0. The order flips inside the band for every arm, so no ranking on counts. IC's group-addition count is a lower bound.

6. Limits

L0 shared VM, one run, public points (reproduction panel). Hosts differ from the 2026-10-16 control (K3300 3.74–40.6, K600 81.5–396), so cross-study numbers are not paired. No replay certificate or validator report: nothing enters the leaderboard or `benchmarks/README.md`.